

Library Management System Case Study

Using Servlets • JSP • Hibernate (Annotations) • HTML • CSS

Case Study Overview

A web application that allows users to register, login and manage library books. Hibernate inheritance mapping is demonstrated through Book, PrintedBook and EBook entities.

Problem Statement

Develop a Library Management System with User Registration/Login and CRUD operations on books using Hibernate annotation configuration and inheritance mapping.

Hibernate Inheritance

Parent: Book

Child 1: PrintedBook

Child 2: EBook

Recommended Strategy: @Inheritance(strategy = InheritanceType.JOINED)

Project Folder Structure

LibraryManagementSystem/

src/

```
├── model/
│   ├── User.java
│   ├── Book.java
│   ├── PrintedBook.java
│   └── EBook.java
├── dao/
│   ├── HibernateUtil.java
│   ├── UserDAO.java
│   └── BookDAO.java
├── servlet/
│   ├── RegisterServlet.java
│   ├── LoginServlet.java
│   ├── LogoutServlet.java
│   ├── AddBookServlet.java
│   ├── ViewBooksServlet.java
│   ├── EditBookServlet.java
│   ├── UpdateBookServlet.java
│   └── DeleteBookServlet.java
```

```
├── util/
│   └── DBConstants.java
└── resources/
    └── hibernate.cfg.xml
```

WebContent/

```
├── css/
│   └── style.css
├── images/
├── index.jsp
├── register.jsp
├── login.jsp
├── dashboard.jsp
├── addBook.jsp
├── editBook.jsp
├── viewBooks.jsp
├── success.jsp
└── error.jsp
```

Screen Designs

1. Home Page

LIBRARY MANAGEMENT SYSTEM

[Register] [Login]

2. Registration Page

User Registration

Full Name

Email

Mobile

Password

Confirm Password

[Register]

Already Registered? Login

3. Login Page

User Login

Email

Password

[Login]

4. Dashboard

Welcome Admin/User

Add Book

View Books

Logout

5. Add Book

Add New Book

Book Type: Printed / EBook

Title

Author

Publisher

ISBN

Price

Category

Printed Book:

Pages

Shelf No

Edition

EBook:

File Format

File Size

Download Link

[Save]

6. View Books

ID	Title	Author	Type	Price	Actions

1	Java	James	Printed	500	Edit Delete
2	HTML	Mark	EBook	250	Edit Delete

7. Edit Book

Edit Book Details
(Update all fields)

[Update]

8. Delete Confirmation

Are you sure you want to delete this book?

[Yes] [No]
